TASK 1 | MAIN ELECTRICAL DISTRIBUTION SCHEMATIC

A3 LANDSCAPE | NOT TO SCALE | 2 x 1500 kVA | 380/220 V LV DESIGN BASIS

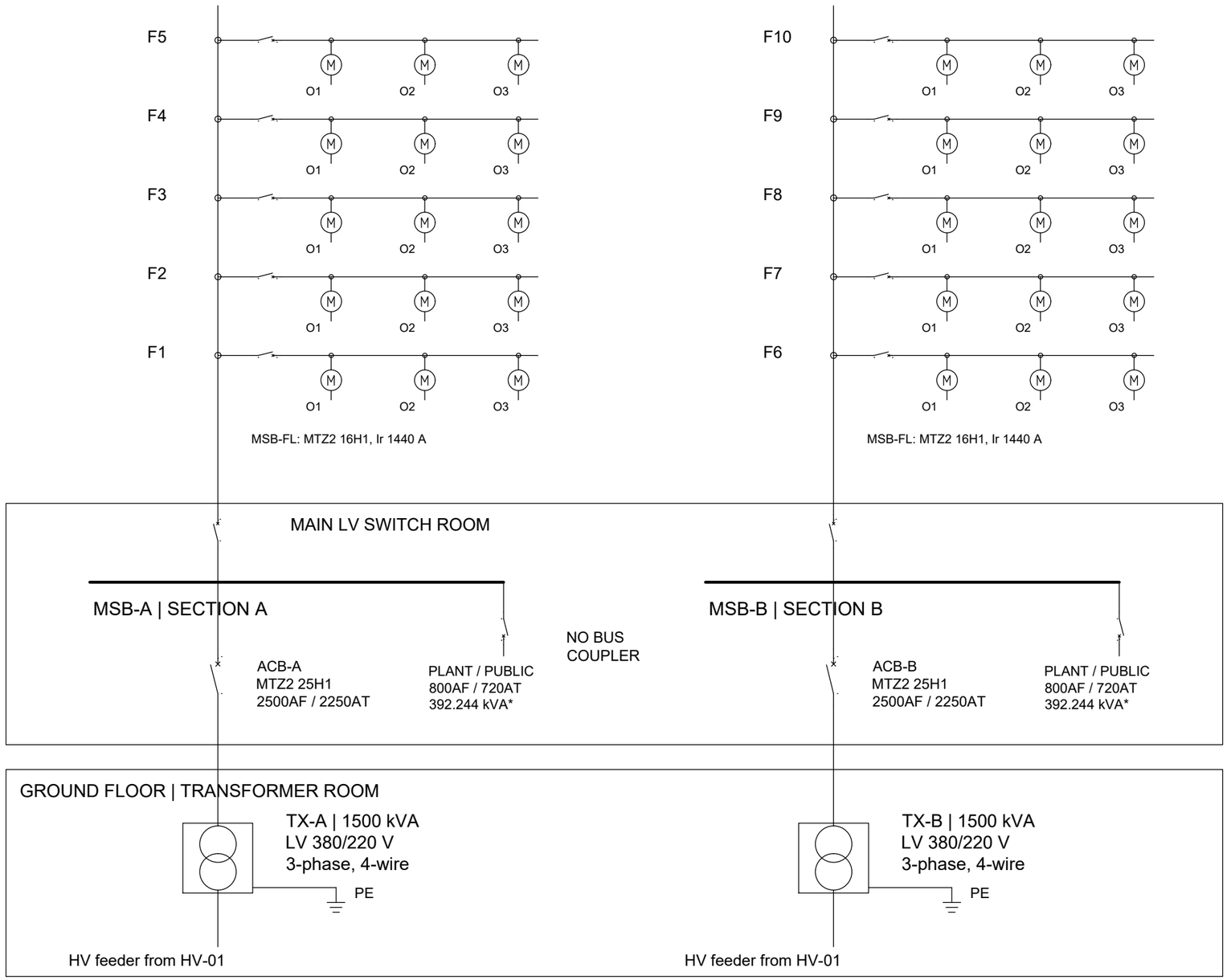
NORMAL SUPPLY: 2672.216 kVA total; two independent LV sections.
Equal allocation: 1336.108 kVA / approx. 2030 A per section.

ZONE A | FLOORS 1-5

1600 A rising main; floor MCCBs 400AF / 320AT

ZONE B | FLOORS 6-10

1600 A rising main; floor MCCBs 400AF / 320AT



* Per section: AC 292.244 + public services 100 = 392.244 kVA before the 1.1 zoning factor.

SYSTEM BASIS

MSB design PFC 51.67 kA: TX uk=6%, -10% tolerance, c=1.1, plus 8 x plant/public current. Independent sections. Main/riser ACBs 66 kA; floor/plant MCCBs 70 kA. AF=frame; AT=long-time setting. Utility supply and protection follow COP101.

REFERENCE: Electrical Schematic 1 - transformer, main switchboard and floor-metering arrangement.
Read with the plantroom layout and Office 1 distribution-board schematic.

TYPICAL OFFICE 1 SUPPLY

Selected tenant detail; applicable to its floor supply zone.

OFFICE 1 | FROM ITS FLOOR METER

M

Office 1 tenant meter

MS-O1 | C63, 4P
A9F54463 | 10 kA

4C35 mm2 PVC Cu
+ CPC16 mm2
31.512 m design length

DB-O1 | 100 A, 4P isolator
15 outgoing RCBO circuits
7 lighting + 8 small power
Ib(max phase) = 48.600 A

P7: WH01 via FS01
SHC10 GB | 1.4 kW
B10 / 10 A DP FCU

TN-S; dedicated N and PE. PE bypasses switching.
Design source: Ze = 0.20 ohm; PFC <=6 kA at MS-O1; upstream drop allowance <=2.2 V.

DESIGN LIMITS

No full-load N-1. Central plant/public load is split equally as a design assumption. This sheet shows normal supply; emergency systems are separate.

GROUP 1 | E-04 | REV 02 | 10 NOV 2026